

Tips Calculation Logic

Developer Documentation

This document defines the **exact rules and formulas** to calculate tips distribution per location and per shift.

This logic must be treated as **financial logic**: deterministic, auditable, and reproducible.

1. High-Level Concept

- Tips are calculated **per location**
 - Each location has **two daily tip pools**:
 - **AM Tips**
 - **PM Tips**
 - Tips are distributed based on **worked hours**
 - Employees may receive tips from **both AM and PM pools** if their shift overlaps
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2. Tip Inputs (Manual)

For each **location and date**, the manager inputs:

- **AM_TIPS_GROSS**
- **PM_TIPS_GROSS**

These are the **gross tip amounts collected** for each shift.

3. Production Pool Deduction

Before any distribution:

- **4% of gross tips** is deducted from each shift
- This amount is allocated to the **Production Pool**

Formula

PRODUCTION_DEDUCTION = **GROSS_TIPS** × 0.04

DISTRIBUTABLE_TIPS = **GROSS_TIPS** - **PRODUCTION_DEDUCTION**

This calculation is applied **separately** for AM and PM.

4. Shift Definitions (Fixed Rules)

AM Shift

- Time range: **06:00 – 15:00**

PM Shift

- Time range: **15:00 – 23:00**

Worked hours must be **split across these time boundaries**, regardless of scheduled shift name.

5. Worked Hours Calculation

Worked hours are derived from **clock-in / clock-out times**.

Per Employee, Per Day:

1. Calculate total worked time
2. Split worked time into:
 - **AM_WORKED_HOURS**
 - **PM_WORKED_HOURS**

Example

Employee works: **09:00 – 17:00**

- AM: 09:00 – 15:00 → **6.0 hours**
 - PM: 15:00 – 17:00 → **2.0 hours**
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6. Total Worked Hours per Shift (Location-Level)

For each location and date:

TOTAL_AM_HOURS = Σ **AM_WORKED_HOURS** (all employees)

$TOTAL_PM_HOURS = \Sigma PM_WORKED_HOURS$ (all employees)

These totals are used to calculate the **Tip Rate**.

7. Tip Rate Calculation

Tip Rate represents the **value of one worked hour** for a given shift.

Formula

$TIP_RATE = DISTRIBUTABLE_TIPS \div TOTAL_WORKED_HOURS$

Calculated separately for:

- AM shift
- PM shift

Example

$AM_DISTRIBUTABLE_TIPS = \320

$TOTAL_AM_HOURS = 80$

$AM_TIP_RATE = 320 \div 80 = \4.00 per hour

8. Employee Tip Allocation

For each employee:

AM Tips

$EMPLOYEE_AM_TIPS =$
 $AM_WORKED_HOURS \times AM_TIP_RATE$

PM Tips

$EMPLOYEE_PM_TIPS =$
 $PM_WORKED_HOURS \times PM_TIP_RATE$

Total Employee Tips

EMPLOYEE_TOTAL_TIPS =
EMPLOYEE_AM_TIPS + EMPLOYEE_PM_TIPS

9. Full Worked Example

Inputs (Single Location, Single Day)

Manager Input

- AM Gross Tips: \$400
- PM Gross Tips: \$600

Production Deduction

- AM: $\$400 \times 4\% = \$16 \rightarrow \$384$ distributable
 - PM: $\$600 \times 4\% = \$24 \rightarrow \$576$ distributable
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Worked Hours

Employee	Clock In	Clock Out	AM Hours	PM Hours
Ivy	09:00	17:00	6.0	2.0

Assume

- Total AM hours (all staff): 96
 - Total PM hours (all staff): 144
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Tip Rates

AM_TIP_RATE = $384 \div 96 = \$4.00$
PM_TIP_RATE = $576 \div 144 = \$4.00$

Ivy's Tip Allocation

AM Tips = $6 \times 4 = \$24$
PM Tips = $2 \times 4 = \$8$
TOTAL = $\$32$

10. Guardrails & Rules

- If `TOTAL_WORKED_HOURS` = 0 → Tip rate = 0 (no distribution)
 - Employees only earn tips for **hours actually worked**
 - No rounding during calculations
 - Final amounts rounded **only at output stage** (2 decimals)
 - Calculations are **per location, per day**
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11. Audit Requirements

System must allow:

- Viewing raw inputs (gross tips, worked hours)
- Viewing intermediate values (tip rates)
- Viewing final employee payouts
- Reproducing results exactly from stored data

Phase 2 — Weekly Tips Payout, Deductions & Production Pool

Developer Documentation (Locked)

This phase builds **on top of Phase 1 (daily calculations)**.
No daily logic may be modified in this phase.

1. Phase 2 Objective

Convert **daily accrued tips** into a **weekly payable amount**, applying:

1. Weekly aggregation (Mon–Sun)
2. Tardiness-based deductions (corrected rules)
3. Manual deductions (admin-controlled)
4. Final weekly payout visibility

This phase determines **what gets paid**, not **how tips are earned daily**.

2. Inputs (From Existing Systems)

From Phase 1 (Daily Engine)

- `EMPLOYEE_DAILY_TIPS`

From Tardiness Module (Already Implemented)

- `WEEKLY_TOTAL_TARDINESS_MINUTES`

From Admin Input

- `MANUAL_DEDUCTION_AMOUNT` (weekly, optional)
 - `MANUAL_DEDUCTION_REASON` (required if amount > 0)
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3. Weekly Aggregation (Base Amount)

For each **employee, per location, per week (Monday–Sunday)**:

Weekly Gross Tips

`WEEKLY_GROSS_TIPS =`
`Σ EMPLOYEE_DAILY_TIPS`

- This value is **system-generated**
 - Not editable
 - Basis for all deductions
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4. Weekly Tardiness Deduction Rules (Authoritative)

Weekly Tardiness Minutes	Deduction %
0 – 5 minutes	0%
> 5 – 10 minutes	15%
> 10 minutes	20%

Boundary Rules (Explicit)

- 5 minutes → 0%
- 10 minutes → 15%
- 10.01+ minutes → 20%
- One tier per week
- No stacking

Determination Logic

```
IF tardiness_minutes <= 5:  
    deduction_percent = 0.00  
ELSE IF tardiness_minutes <= 10:  
    deduction_percent = 0.15  
ELSE:  
    deduction_percent = 0.20
```

5. Tardiness Deduction Calculation

`TARDINESS_DEDUCTION_AMOUNT =`
`WEEKLY_GROSS_TIPS × deduction_percent`

`WEEKLY_AFTER_TARDINESS =`

WEEKLY_GROSS_TIPS - TARDINESS_DEDUCTION_AMOUNT

6. Manual Deduction Layer (Admin-Controlled)

Manual deductions are applied **after tardiness**.

Rules

- Fixed amount only (no percentages)
- Cannot exceed remaining balance
- Reason field mandatory
- Fully auditable

Calculation

WEEKLY_AFTER_MANUAL =
WEEKLY_AFTER_TARDINESS - MANUAL_DEDUCTION_AMOUNT

NET_WEEKLY_TIPS =
 $\max(0, \text{WEEKLY_AFTER_MANUAL})$

Net tips **cannot be negative**.

7. Weekly Staff Payout Table (Required UI)

Per **Employee × Location × Week**:

Field	Description
Employee	Name
Location	Location worked
Daily Tips From Mon - Sun	Total of daily tips
Weekly Tardiness (min)	From tardiness module
Tardiness %	Applied tier
Tardiness Deduction	System-calculated

Manual Deduction Admin input
Net Weekly Tips Payable Final amount

8. Processing Order (Locked)

1. Daily staff tip accrual (Phase 1)
2. Weekly aggregation
3. Tardiness deduction
4. Manual deduction
5. Net weekly staff payout

Any deviation breaks traceability.

9. Audit & Traceability Requirements

System must allow:

- Drill-down: weekly → daily → shift
 - Visibility into:
 - Tardiness tier selection
 - Manual deduction reason
 - Recalculation of any historical week with identical results
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Final Lock Statement (For Developer)

This is a **staff-only tip payout system**.

- Tardiness impacts **tips**
- Manual deductions are **exceptions**
- All math must be **deterministic and auditable**

Any assumption or shortcut must be **documented and approved before implementation**.

Phase 2 — Addendum

Tardiness Deduction Redistribution Logic (Staff Tips Only)

This section defines how **tardiness deductions are redistributed to the rest of the team** at the end of the week.

This logic is **additive** and does **not alter** how deductions are calculated.

1. Concept Overview

- Tardiness deductions are **not removed from the system**
 - Deducted amounts are **redistributed to punctual staff**
 - Redistribution happens **after all weekly deductions are finalized**
 - Only **eligible staff** participate in redistribution
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2. Redistribution Pool Definition

For each **location**, **per week**:

$$\text{REDISTRIBUTION_POOL} = \sum (\text{TARDINESS_DEDUCTION_AMOUNT})$$

This pool represents the **total tips withheld due to tardiness**.

3. Eligibility Rules

An employee is **eligible to receive redistributed tips** if:

- Weekly tardiness $\leq$ **5 minutes**
- Employee has **worked hours > 0** in the week
- Employee is not fully deducted (i.e. did not trigger a tardiness deduction)

Employees who received a tardiness deduction:

- **Do NOT** receive any portion of the redistribution pool

4. Redistribution Basis

Redistribution is based on **worked hours**, not equal split.

Per Location, Per Week:

$$\text{ELIGIBLE_TOTAL_HOURS} = \sum \text{EMPLOYEE_WEEKLY_WORKED_HOURS}$$

(only eligible employees)

5. Redistribution Calculation (Per Employee)

For each eligible employee:

$$\begin{aligned} \text{EMPLOYEE_REDISTRIBUTED_TIPS} = & \\ & (\text{EMPLOYEE_WEEKLY_WORKED_HOURS} \div \text{ELIGIBLE_TOTAL_HOURS}) \\ & \times \text{REDISTRIBUTION_POOL} \end{aligned}$$

6. Final Weekly Payout Adjustment

After redistribution:

$$\begin{aligned} \text{FINAL_WEEKLY_TIPS} = & \\ & \text{NET_WEEKLY_TIPS} \\ & + \text{EMPLOYEE_REDISTRIBUTED_TIPS} \end{aligned}$$

This value becomes the **final payable amount**.

7. Display & Transparency Requirements

Weekly payout table must now include **two additional fields**:

Field	Description
Tardiness Redistribution	Tips received from others

Final Weekly Tips Payable Net tips + redistribution

Employees with tardiness deductions will show:

- Redistribution = \$0.00
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8. Guardrails (Non-Negotiable)

- Redistribution is **location-specific**
 - Redistribution pool must be **fully distributed**
 - No redistribution if pool = 0
 - Redistribution does **not override** manual deductions
 - Redistribution does **not reduce** anyone else's net tips
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9. Processing Order (Extended)

Updated final steps only:

1. Daily staff tip accrual
 2. Weekly aggregation
 3. Tardiness deduction
 4. Manual deduction
 5. Net weekly tips
 6. **Tardiness redistribution pool creation**
 7. **Redistribution to eligible staff**
 8. Final weekly staff payout
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Final Lock Statement (For Developer)

This redistribution layer is **behavioral reinforcement**, not penalty.

- Late staff lose tips
- Punctual staff gain tips
- Total staff tip pool remains constant

If this redistribution is skipped or implemented incorrectly, **the incentive system fails.**